



Numerical Analysis on Aerodynamic Characteristics of Slender Body with Asymmetric Double Protuberances

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In order to achieve safe flight of the flight vehicles, the relationship between the arrangement of protuberances attached to the fuselage and the aerodynamic characteristics has been investigated. Kawauchi et al. (2019) have investigated the side force characteristics of a slender body equipped with a single protuberance. However, the relationship between the arrangement of the multiple protuberances and the side force characteristics was not studied enough. Therefore, in this study, we conducted numerical calculations for supersonic flow around the slender body with double protuberances using the supercomputer in JAXA. According to the results, it is revealed that, when one protuberance is attached to the leeward of the front portion of the vehicle and the other to the leeward of the middle portion, the side force is not the sum of the single protuberance cases and is about 2% less than the value of the case with single protuberance at the front portion of the vehicle. (Note that both protuberances are mounted on the port side of the vehicle.) This is because the behavior of the vortices originating from the protuberance at the middle portion is different between the single protuberance case and the double protuberance case (single protuberance case; merges with other vortices, double protuberance case; exists independently of other vortices).

Nomenclature

a	=	speed of sound
C_p	=	pressure coefficient
C_Y	=	side force coefficient
D	=	diameter of the vehicle
e_T	=	total energy
F_y	=	y component of the pressure force acting on the vehicle
h_p	=	height of the protuberance
L	=	total length of the vehicle
l_p	=	length of the protuberance
M	=	Mach number
p	=	pressure

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Pr	=	Prandtl number
q	=	dynamic pressure
Re	=	Reynolds number
S	=	reference area based on D of the vehicle
T	=	temperature
U	=	freestream velocity
u	=	x component of velocity
v	=	y component of velocity
w	=	z component of velocity
α	=	angle of attack
γ	=	specific heat ratio
δ	=	Kronecker's delta
κ	=	thermal conductivity
μ	=	molecular viscosity
ρ	=	density
τ	=	stress tensor
φ	=	azimuthal angle of the protuberance

Subscripts

0	=	stagnation point condition
∞	=	free stream quantity
c_g	=	center of gravity
t	=	turbulence quantity

I. Introduction

In recent years, the development of small satellites or nanosatellites has been actively pursued [1,2]. With the future development of IoT (Internet of things) technology and information technology, the demand for communication devices, including mobile phones, is expected to increase further [3], and the number of satellite launches by small space transporters, which have the advantage of low cost, is on the rise [1,2]. However, the downsizing of the vehicle limits its internal volume. Thus, various protuberances (feed pipes, cable ducts, auxiliary devices, cameras, etc.) must be attached to the outside of the body in order to ensure as much payload as possible. These protuberances have various effects on the aerodynamic characteristics of the vehicles. For example, in supersonic flow, which is the main flight condition for rockets, shock waves formed upstream of the protuberances create wave drag, which worsens the fuel consumption of the rocket [4]. The vortices created in the wake of the protuberance may cover the control devices of the flight vehicle, such as the sub-thrust motor, which may prevent the desired flight performance or make it difficult to control the attitude of the vehicle [5]. It is also known that protuberances on the surface of the vehicle have a significant effect on the side force characteristics related to the yaw control. If the protuberances are mounted asymmetrically to the flight vehicle, the vortices generated around the vehicle are also asymmetrical, resulting in the asymmetric pressure distribution on the body surface. Therefore, the side force is generated in the lateral direction with respect to the travel direction [6]. The side force may prevent the flight vehicle from flying in its intended path. Hence, for a safe and successful mission, the relationship between the side force and the flow physics around the vehicle (e.g., vortices and shock waves) must be clarified, and the proper placement, number, and shape of protuberances must be investigated.

Many investigations have been conducted on the relationship between the surface protuberance and the side force. Kawauchi et al. [6] investigated the effect of the arrangement of single protuberance on the side force for the slender-bodied vehicle with a slenderness ratio of 8.9. As a result, it was clarified that the side force increases even at a relatively low angle of attack ($\alpha = 15^\circ$) when the protuberance is attached to the leeward side under the fairing (Fig. 1). However, these results are limited to the case of a single protuberance. On an actual flying vehicle, multiple protuberances are often placed on the surface. For example, in the Ares I rocket, there are many protuberances on the surface of the vehicle, such as the roll control system, reaction control system, liquid hydrogen (LH2) feedline fairing, system tunnel, booster deceleration motor (BDM), cameras, and antennas [7]. In such cases where multiple protuberances are attached to the vehicle surface, a more complex flow field is expected to form than in the single protuberance case. However, most of the investigations have been conducted for specific projects [7-10] and have not yet been systematically summarized.

In the previous study [6], it was revealed that the behavior of the wake vortex of the protuberance is dominant for the side force acting on the vehicle. In other words, even if it is necessary to attach multiple protuberance to the body surface in the process of designing, it may be possible to reduce the side force as much as possible by predicting the behavior of the wake vortices and placing protuberances in appropriate positions. Thus, the purpose of this study is to help achieve efficient and safe missions in terms of the relationship between the location of protuberances and aerodynamic forces.

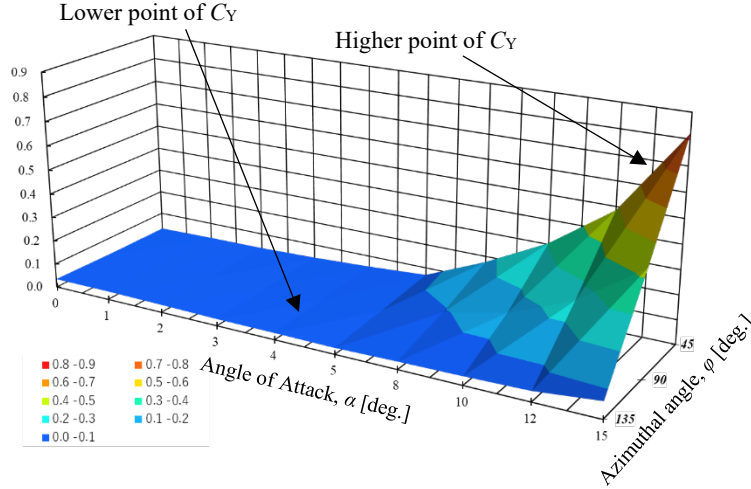


Fig. 1 Side force coefficient C_Y vs angle of attack and azimuthal angle of the protuberance [6].
(Note that the protuberance is mounted at the front portion of the vehicle,
and the larger the azimuthal angle, the more the protuberance is located leeward.)

II. Numerical Setup

A. Flight Vehicle Configuration for Analysis

The flight vehicle (clean) configuration for this analysis is a simple slender-bodied rocket which is composed of an axisymmetric body (hemispherical cone + cylinder), shown in Fig. 2a. The total length of the vehicle $L = 368.0\text{mm}$, the diameter of the cylinder $D = 41.5\text{mm}$, and the fineness ratio L/D is about 8.9. The reference area is the base area of the cylinder $S (= \pi D^2/4)$. Besides, the center of gravity x_{cg} is about 78% of L . Protuberances will be attached to this clean configuration.

The coordinates are the Cartesian coordinate system of the right-handed system (x, y, z) , with the nose tip of the body as the origin, the x -axis is along the body axis, the y -axis is the rotation axis direction where the angle of attack α is defined, and the z -axis is defined to be perpendicular to each of the two axes.

The protuberance imitates the size of the sub-thrust motor cover of the Epsilon rocket [8-10]; the protuberance height is $h_p = 0.15D$, and the length is $l_p = 0.06L$ (Fig. 2b). Two protuberances are arranged asymmetrically with respect to an axisymmetric body, and the positions of the protuberances are changed in various ways. The positions of the protuberances are shown in Fig. 3. We vary the positions of the protuberance as follows; i) Front; $x/L = 0.22$, and ii) Middle; $x/L = 0.55$. The reason why we chose the two positions is that protuberances are typically seen on those positions in general rockets [11]. Figure 3b presents the azimuthal positions of the protuberance, where φ denotes the azimuthal angles as measured clockwise from the z -axis in the direction of the x -axis. In this study, there are three azimuthal locations for each axial position; $\varphi = 45^\circ, 90^\circ$, and 135° . In other words, double protuberances are attached always on the negative y side (port side) of the vehicle (Note that the windward side of the symmetrical plane is 180° and the leeward side is 0°).

In this study, we refer to the protuberance positions of each case as follows for the sake of explanation. For example, a case in which one protuberance is located at the axial position “Front” and the azimuthal angle $\varphi = 45^\circ$, and another at the axial position “Middle” and the azimuthal angle $\varphi = 90^\circ$ will be coined as “F+45_M+90”. Figure 4 shows the models of the four cases that are the main objects of discussion in this study.

B. Governing Equations

In this study, the governing equations are three-dimensional compressible Navier-Stokes equations (1a),

$$\frac{\partial \mathbf{Q}}{\partial t} + \frac{\partial \mathbf{F}_k}{\partial x_k} = \frac{1}{Re} \frac{\partial \mathbf{F}_v}{\partial x_k} \quad (1a)$$

where $\mathbf{Q}$ is the conservation vector, $\mathbf{F}_k$ is the inviscid flux vector, and $\mathbf{F}_v$ is the viscous flux vector. These are described as follows: (1b-c).

$$\mathbf{Q} = \begin{pmatrix} \rho \\ \rho u_l \\ e_T \end{pmatrix}, \quad \mathbf{F}_k = \begin{pmatrix} \rho u_k \\ \rho u_l u_k + p \delta_{lk} \\ \rho u_k H \end{pmatrix}, \quad \mathbf{F}_v = \begin{pmatrix} 0 \\ \tau_{lk} \\ u_m \tau_{mk} + \frac{\kappa}{(\gamma-1)Pr} \frac{\partial T}{\partial x_k} \end{pmatrix} \quad (1b)$$

$$\tau_{lk} = \mu \left(\frac{\partial u_l}{\partial x_k} + \frac{\partial u_k}{\partial x_l} \right) - \frac{2}{3} \mu \frac{\partial u_n}{\partial x_n} \delta_{lk} \quad (1c)$$

where subscripts k, l, m , and n take 1, 2, 3 denoting the Cartesian coordinates, ρ is density, u_i is velocity components ($i = 1, 2$, and 3 corresponds to u, v , and w , respectively), e_T is total energy, p is pressure, T is temperature, and all quantities are dimensionless. The working gas is air approximated by the calorically perfect gas model with the specific heat ratio $\gamma = 1.4$. The Prandtl number is $Pr = 0.71$. The molecular viscosity μ is calculated from Sutherland's law. The thermal conductivity κ is obtained from $\kappa = \mu$, assuming that Prandtl number is constant with $Pr = 0.71$. For modeling turbulence, the molecular viscosity μ is replaced by $(\mu + \mu_t)$, where μ_t is the turbulence viscosity given by the Spalart-Allmaras model. Likewise, κ is replaced by $(\mu + \mu_t Pr / Pr_t)$, where the turbulent Prandtl number $Pr_t = 0.90$. Reynolds number is defined as Eq. (2).

$$Re = \frac{\rho_\infty U_\infty L a_\infty}{\mu_\infty U_\infty} = \frac{Re_\infty}{M} \quad (2)$$

where a is speed of sound, U is freestream of velocity, M is Mach number ($M = U/a$).

C. Computational Methods

In this study, we performed numerical calculations using a 3D high-speed flow solver for unstructured grids, FaSTAR [12] developed at the Japan Aerospace Exploration Agency (JAXA). The system of equations (Eq. (1)) is discretized by cell-centered finite volume method. We use Spalart-Allmaras One-Equation Model without a numerical fix term (SA-nof2-R) [13] as the turbulence model. A numerical flux for an inviscid term is calculated by an advection upstream splitting method (AUSM) type scheme, simple low-dissipation AUSM (SLAU) [14]. Green-Gauss method [15] is used to reconstruct the gradient and a slope limiter is minmod [16]. For the time integration method, lower-upper symmetric Gauss-Seidel scheme [17] is adopted. The adoption of these methods results in a second-order accuracy in both spatial and temporal terms. The present calculations assumed a steady flow condition.

D. Computational Grids

We use HexaGrid [18], a hexahedron-based automatic grid generation tool that creates computational grid from CAD geometry data in the STL format. The computational grid of F+45_M+45 case is shown in Fig. 5a. The coordinate system is defined as mentioned above. As shown in Fig. 5b, the computational domain is a $20L \times 20L \times 20L$ cube, composed of hexahedra, prisms, pyramids, and tetrahedra. The grid width of the first layer of the wall is set to 1.0×10^{-3} mm ($y^+_{\max} = 0.86$). In order to confirm the grid convergence, three types of grids (coarse: 20 million cells, medium: 43 million cells, fine: 86 million cells) were prepared for the F+45_M+45 case, and numerical calculations were performed under the conditions of $\alpha = 15^\circ$ and $M_\infty = 1.5$ (more detailed conditions are described in Sec. II-E). The calculated values of the side force coefficient C_Y (to be defined later in Sec. II-F) are shown in Table 1. The error between the coarse grid and the medium grid was about 6.3%, whereas the error between the medium grid and fine grid was about 1.0%. Therefore, the side force coefficient C_Y of the medium grid is considered to be sufficiently converged, and that was used in this study.

E. Computational Conditions

The computational conditions are selected to correspond to the wind tunnel test conditions [19] in Table 2, such as the freestream Mach number $M_\infty = 1.5$, the Reynolds number Re (based on the length of the body), the total pressure p_0 , total temperature T_0 , and angle of attack α . Mach 1.5 is selected because this is the speed at which a rocket experiences its highest dynamic load (i.e., maximum dynamic pressure q)[6]. This is one of the most important Mach numbers during flight for many rockets. Moreover, in the case with a single protuberance, the maximum side force coefficient C_Y was generated at $\alpha = 15^\circ$ [6]. Therefore, the angle of attack α was selected as $\alpha = 15^\circ$.

F. Aerodynamic Coefficients

We defined the side force coefficient as follows.

$$C_Y = \frac{F_y}{q_\infty S} \quad (3b)$$

$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2 \quad (3c)$$

where F_y is y component force, q_∞ is freestream dynamic pressure, S is reference area (based on base diameter D), ρ_∞ is freestream density, and U_∞ is freestream velocity.

In addition, the pressure coefficient C_p is given as

$$C_p = \frac{p - p_\infty}{q_\infty} \quad (4)$$

where p is local pressure, and p_∞ is freestream pressure.

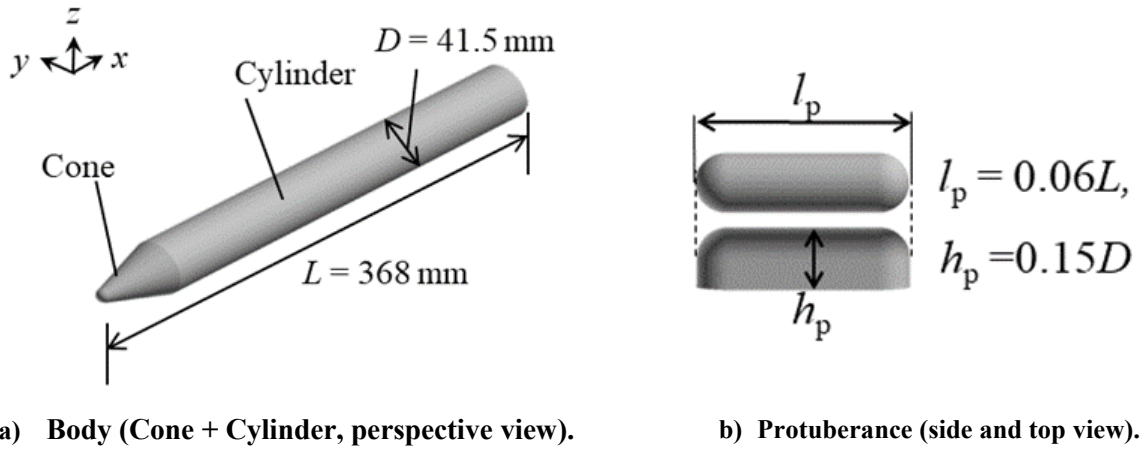


Fig. 2 Configuration.

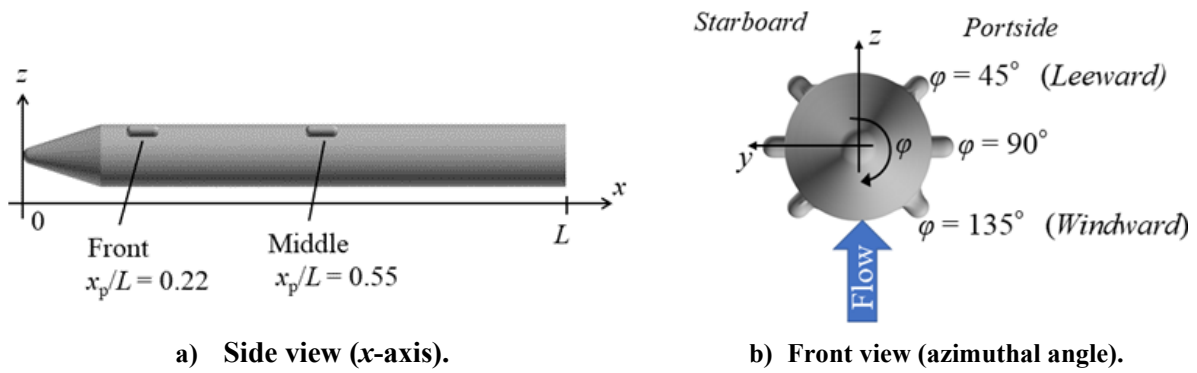


Fig. 3 Protuberance positions.

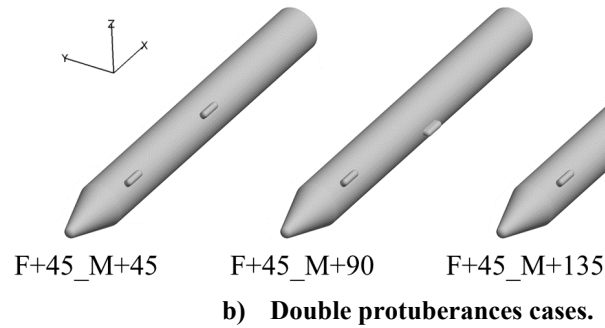
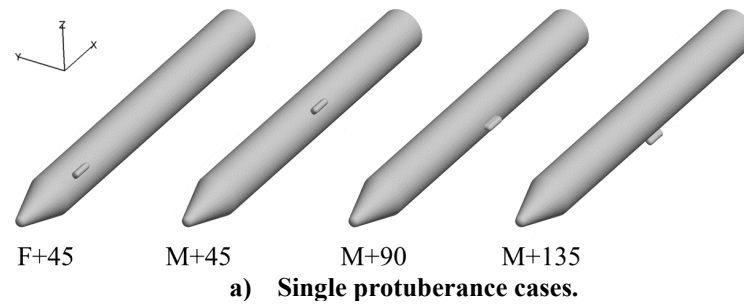


Fig. 4 Caluculated cases.

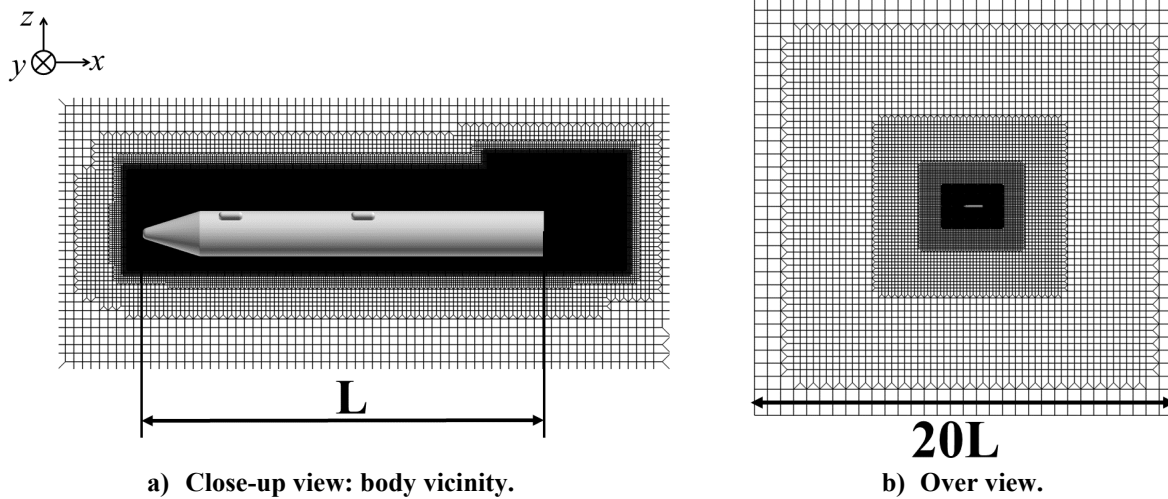


Fig. 5 Computational grid (F+45_M+45).

Table 1 Grid convergence in the F+45_M+45 case.

Grid	Number of Cells	C_Y
Coarse	20 million	0.781
Medium	43 million	0.825
Fine	86 million	0.833

Table 2 Test conditions [19]

M_∞	1.5
Re	1.1×10^7
p_0 , kPa	196.1
T_0 , K	294
α , deg.	15
Run Time, s	15 ~ 35

III. Results and Discussion

A. Validation

At first, the validity of the calculation results is confirmed by comparison with the wind tunnel test [19]. Figure 6 shows the calculated and experimental values of the side force coefficient C_Y for three double protuberance cases. The maximum error between the experimental and calculated values is about 3% (F+45_M+45), and we conclude that the experimental and CFD values are in quantitatively good agreement. Figure 7 shows a comparison of the flow field between CFD and the experiment in the F+45_M+45 case. According to this figure, the computed results capture the detached shock in front of the fairing and the shock waves in front of the double protuberances. The position of the separation vortex at the aft portion of the body and the separation line on the portside are almost identical. Therefore, the CFD results are qualitatively consistent with the experimental results. Based on the above validation, we regard the CFD results as reasonable, and the results of numerical calculations will be used in the following discussion.

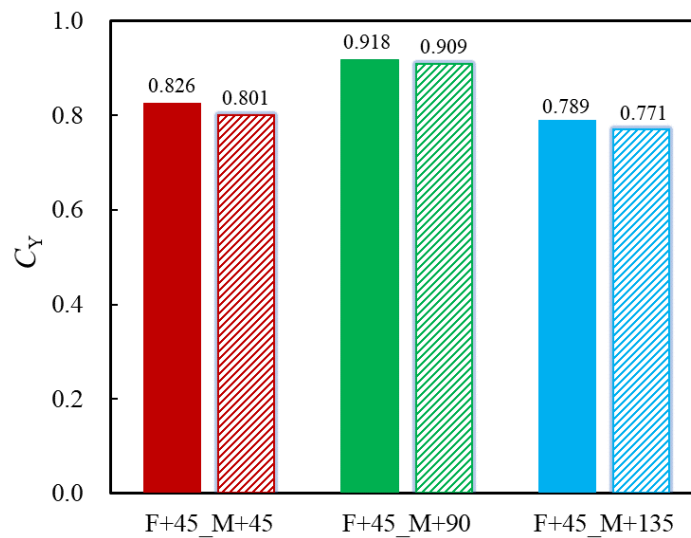


Fig. 6 Calculated and experimental values of the side force coefficient C_Y .
 (Note that the filled bar represents the CFD and the striped bar represents the experimental value.)

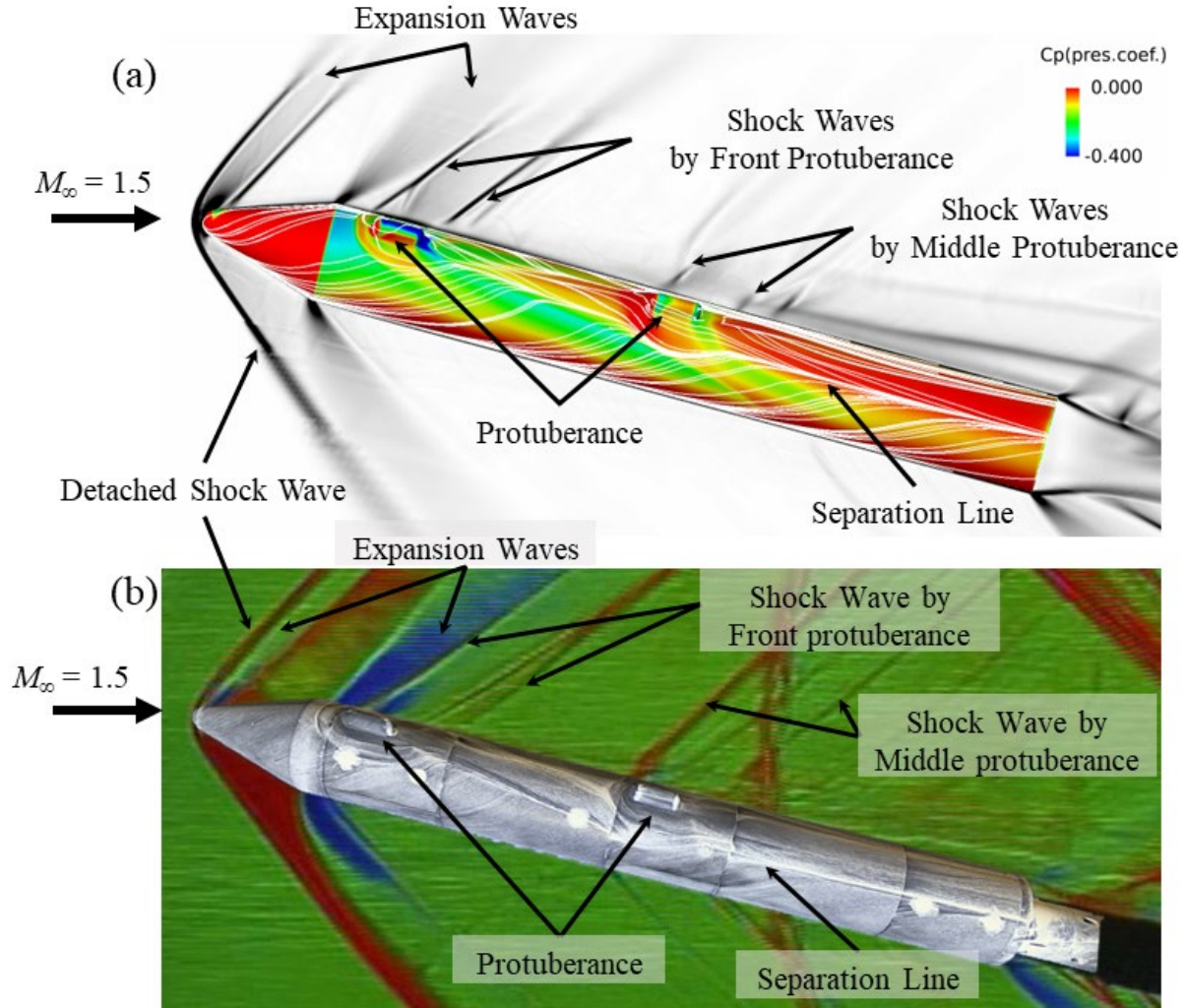


Fig. 7 Flowfield of side view (F+45_M+45 case, $\alpha = 15^\circ$),
 (a) off-surface: density gradient magnitude, on-surface; computed surface C_p and stream-lines.
 (b) WT [19], off-surface: schlieren visualization, on-surface; oil flow image.

B. The Side Force Coefficient for Double Protuberance cases.

In this section, we compare the side force coefficients C_Y for three double protuberance cases (F+45_M+45, F+45_M+90 and F+45_M+135), along with four single protuberance cases (F+45, M+45, M+90 and M+135) numerically obtained in the previous study [6]. For these three double protuberance cases, the second protuberance is added at the middle position to the F+45 case. The reason why we used the F+45 case as the baseline is because it exhibits the maximum side force among single protuberance cases [6].

Figure 8 shows the side force coefficients C_Y for all the cases to be compared. According to the Fig. 8, the side force of F+45_M+135 case is close to the sum of the values of single protuberance cases (F+45 and M+135). In addition, the reason why negative side force is generated in the M+135 case was not investigated in detail. Therefore, M+135 and F+45_M+135 cases are discussed in Sec. IV-B-1. On the other hand, the side force of F+45_M+45 case is not the sum of the values of single protuberance cases (F+45 and M+45). Despite the fact that both the F+45 and the M+45 generate positive side forces, the side force of the F+45_M+45 case is about 1.9% less than that of F+45 case. This is discussed in Sec. IV-B-2.

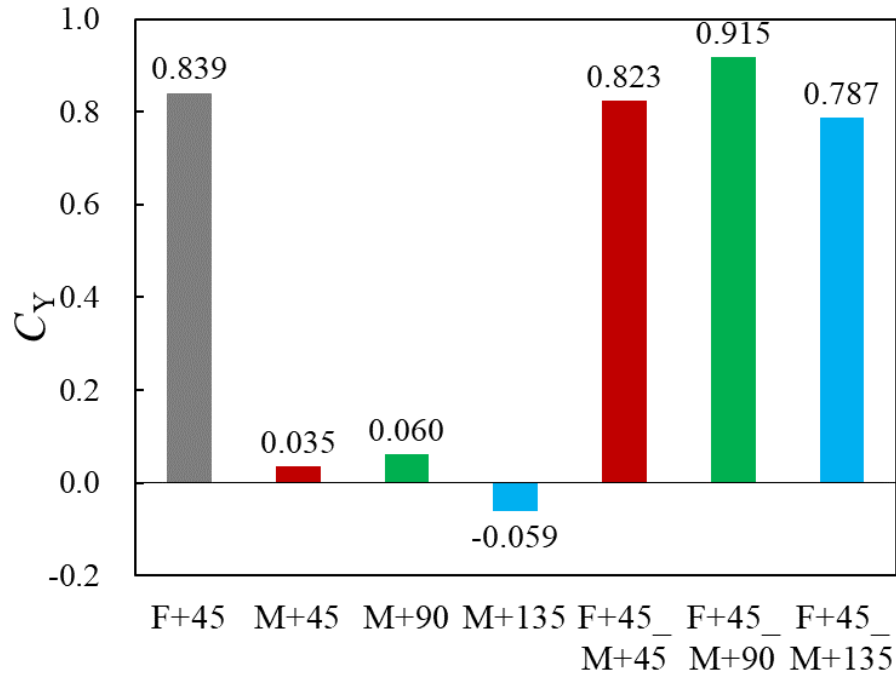


Fig. 8 The side force coefficients C_Y .
(The single protuberance cases are adopted from [6].)

1. M+135 and F+45_M+135

First, we discuss the M+135 case, which generates a negative side force, and F+45_M+135 case, whose side force is nearly the sum of the single protuberance cases (F+45 and M+135). We show in Fig. 9 the local side force distributions [6,10] in order to quantitatively identify where the differences in the side forces occur on the vehicle. According to this figure, for the M+135 and F+45_M+135 cases, the local side force is reduced around the Middle protuberance (referred to as (i) in the figure) and at the aft portion of the vehicle. Moreover, the local side force distributions of the M+135 and the F+45_M+135 cases show a similar behavior behind the Middle protuberance ($x/L > 0.48$). The causes of this tendency are discussed by visualizing the flow field.

(i) Reduction of the Local Side Force around the Middle Protuberance

In this section, we discuss the local side force reduction in region (i) in Fig. 9. Figure 10 shows the C_p distributions on the vehicle surface and the density gradient distributions on the symmetry planes of the Middle protuberances. In Fig. 10, the shock wave S.W.1 is generated in front of the Middle protuberance for both M+135 and F+45_M+135. In addition, the expansion wave E.W.1 is generated at the anterior shoulder of the Middle protuberance. Figure 11 shows the surface C_p distributions and the y - z cross-sectional velocity vectors at the aft portion of the Middle protuberance ($x/L = 0.57$) in order to investigate the effects of these factors on the surface pressure around the Middle protuberance. In this figure, in both M+135 and F+45_M+135 cases, a region of negative pressure by E.W.1 (highlighted by the red dotted line) is formed at the anterior part of the Middle protuberance. In addition, a separation vortex (V4) is formed at the rear of the protuberance, and this V4 forms a negative pressure region (highlighted by the red solid line) on the leeward side of the protuberance.

As described above, two different factors (expansion wave E.W.1 and a separation vortex V4) form a negative pressure region around the Middle protuberance on the port side of the vehicle, which pulls the vehicle in the $-y$ direction (from the side with no protuberances to the side with protuberances), and thus the local side force is significantly reduced in region (i) of Fig. 9. In addition, for the F+45_M+135 case ($C_Y = 0.787$), such a formation of the negative pressure region by the Middle protuberance contributes to a significant reduction in the local side force in the region (i) compared to the F+45 case ($C_Y = 0.839$).

(ii) Reduction of the Local Side Force at the Aft Portion of the Vehicle

Next, we discuss the reduction of the local side force in the region (ii) in Fig. 9. Figure 12 shows the visualization results of the Q iso-surfaces. According to this figure, vortices V4 and V5 are generated downstream of the Middle protuberance in the M+135 and the F+45_M+135 cases. The vortex V4 is a separation vortex from the protuberance (Fig. 11) as explained in Sec. IV-B-1-(i). Figure 13 shows the velocity vector and the vorticity contour line at the $x/L = 0.75$ cross section to clarify the formation mechanism of the vortex V5. This figure shows that a new separation vortex is generated at the aft portion of the vehicle of both M+135 and F+45_M+135, which is confirmed to be V5.

In order to investigate the effects of vortices V4 and V5 on the surface pressure of the vehicles, Fig. 14 shows the surface C_p distributions on the port side of the body and the visualization results of vorticity at the $x/L = 0.9$ cross section. According to this figure, where the vortices V4 and V5 are in close proximity to the body surface, the surface pressure of M+135 and F+45_M+135 is low (highlighted in red in the graph). This is mainly due to the effect of vortex V4, with the effect of V5 as a secondary effect. Due to the above mechanism, in both M+135 and F+45_M+135 cases (with or without the F+45 protuberance), the surface pressure on the port side of the aft portion of the vehicle is lowered and the local side force is reduced in region (ii) of Fig. 9.

As discussed in Sec. IV-B-1-(i), similar phenomena, i.e., formation of negative pressure region by expansion waves and separation vortices, occurred around the Middle protuberance in the M+135 and F+45_M+135 cases. Also, as discussed in (ii), in both the M+135 and F+45_M+135 cases, the separation vortices from the Middle protuberance (V4) and from the body surface (V5) exist independently from the vortices originating from the front portion of the body, and behave in almost the same manner (proximity to the body surface). Therefore, since the local side force distribution behind the front end of the Middle protuberance ($x/L > 0.48$) shows similar behavior, the side force of the F+45_M+135 case is the sum of the single protuberance cases (F+45 and M+135).

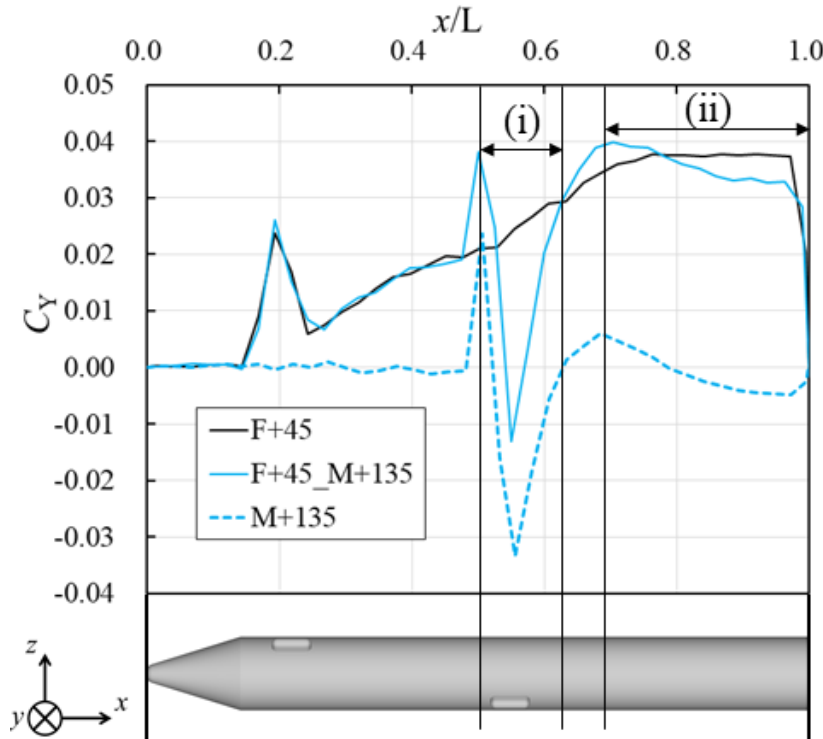


Fig. 9 Local C_y distribution in the axial direction.

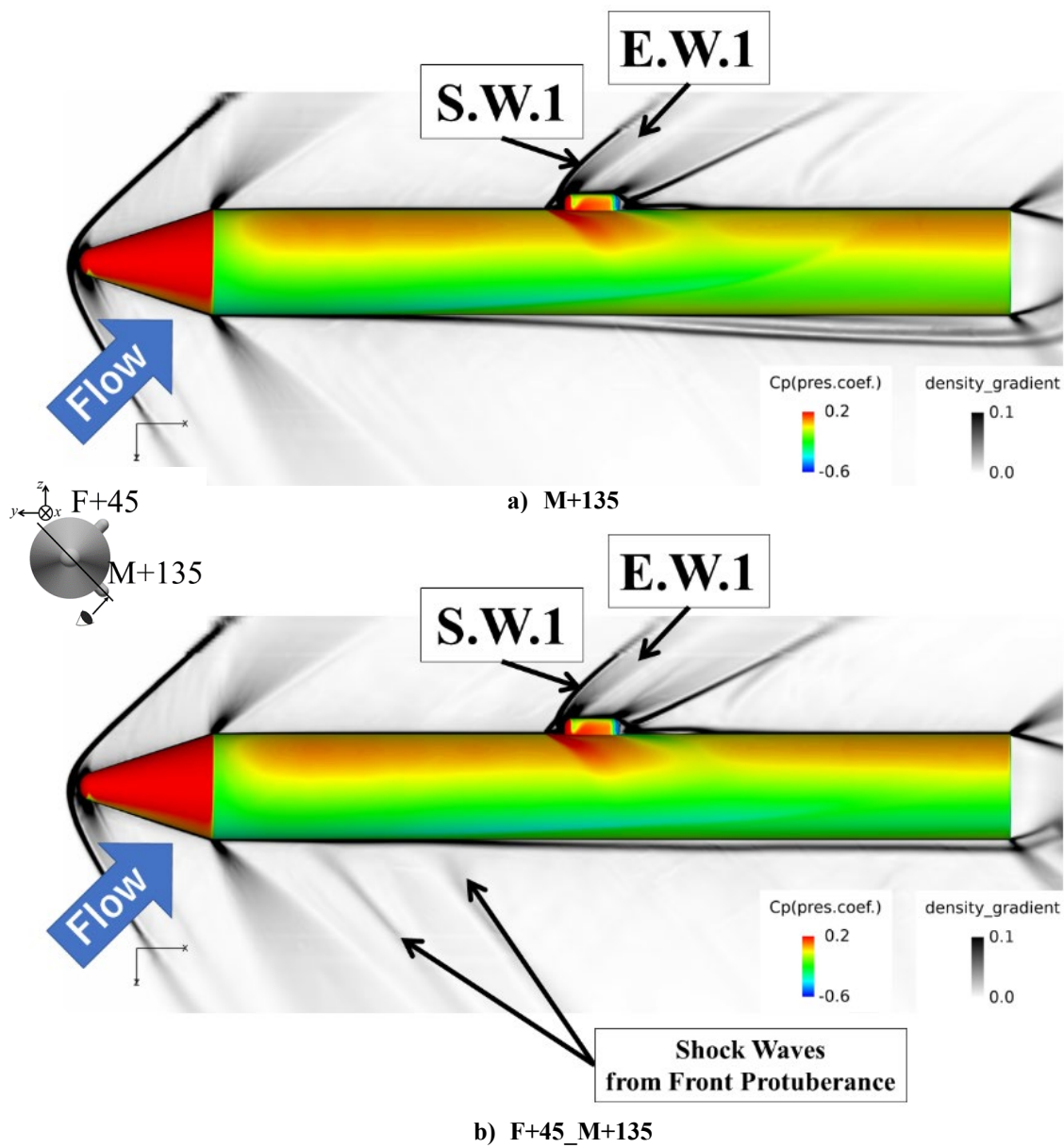
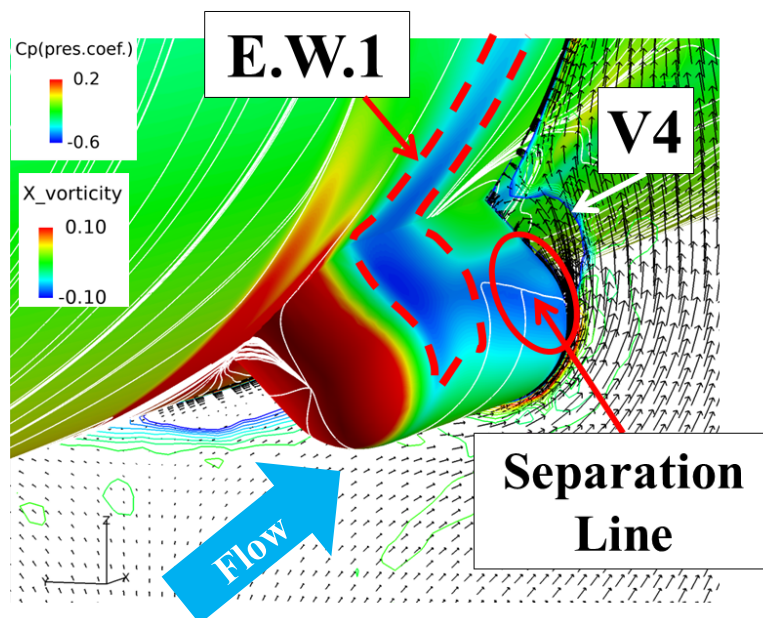
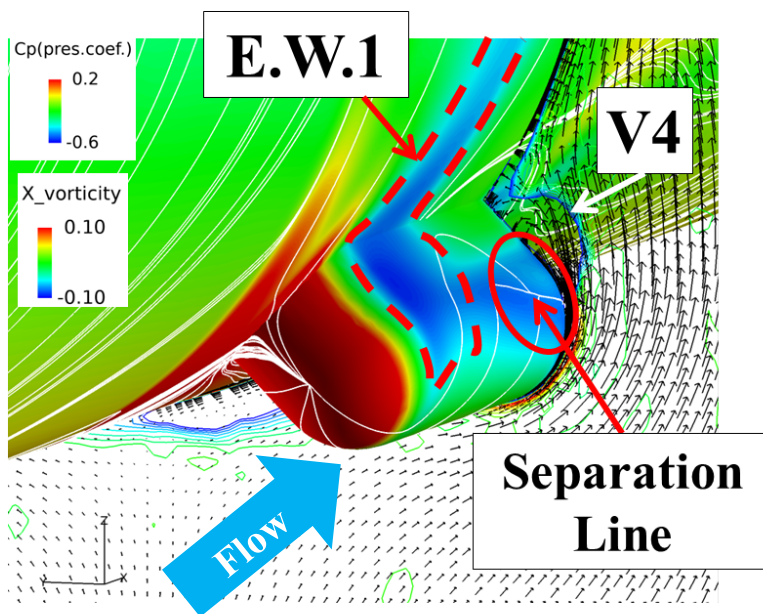


Fig. 10 Surface C_p Distribution and Density Gradient Cross Section.



a) M+135



b) F+45_M+135

Fig. 11 Velocity Vectors and vorticity contours at $x/L = 0.57$ and surface C_p distribution and surface streamlines. (Note that we look at the Middle protuberance from upstream.)

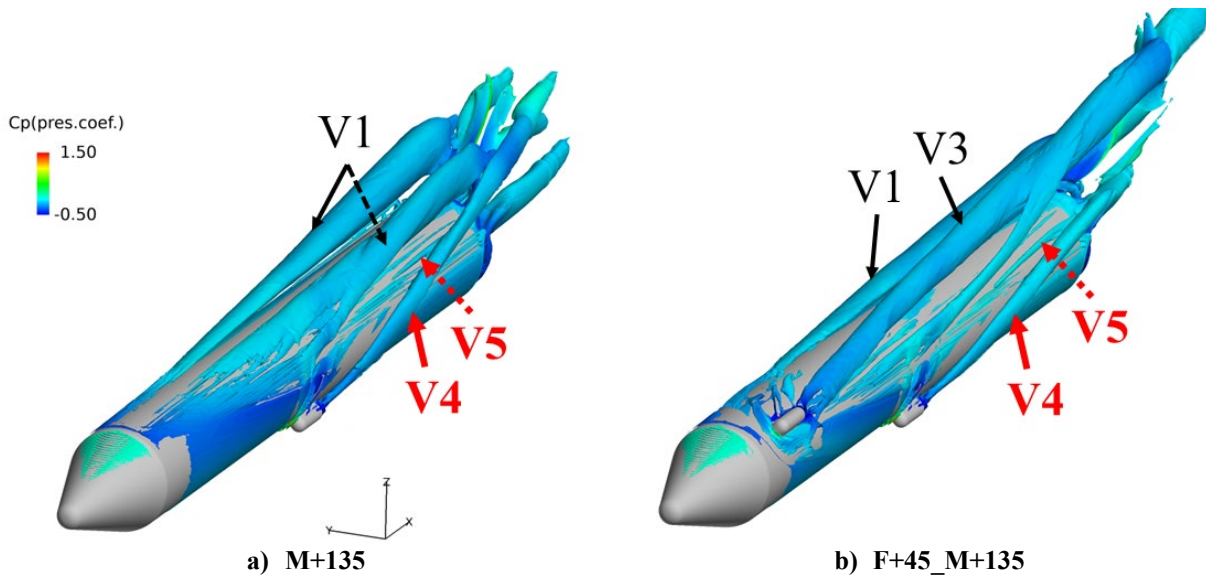


Fig. 12 Q iso-surface (colored with C_p).

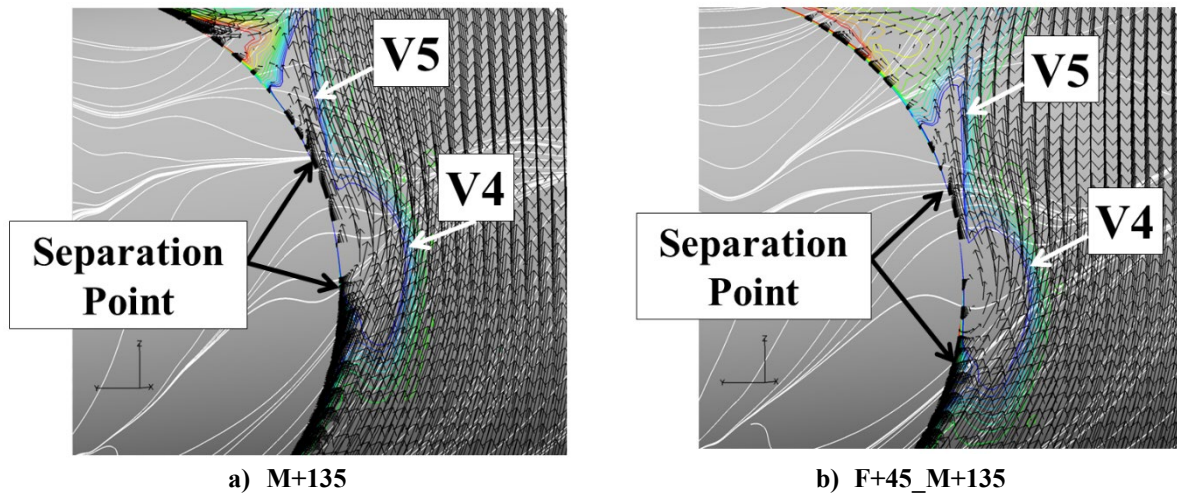


Fig. 13 Velocity vectors and vorticity distributions for the $x/L = 0.75$ cross section and surface streamlines.

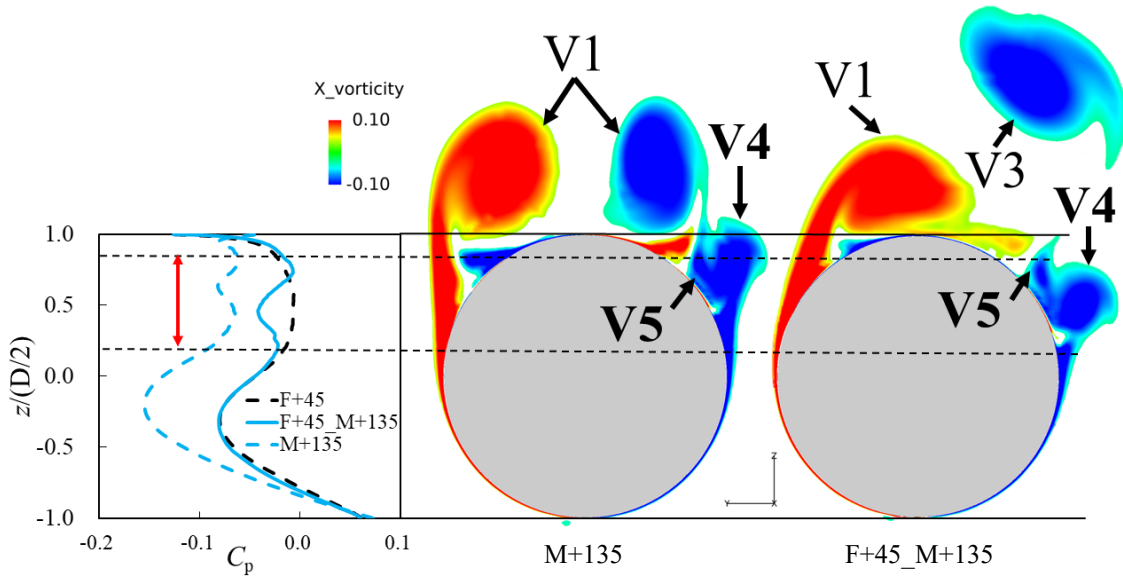


Fig. 14 C_p distributions and two-dimensional (y - z) plane colored with vorticity at $x/L = 0.9$.

2. M+45 and F+45_M+45

Next, the reason why the side force coefficient C_Y of the F+45_M+45 case is not the sum of the single protuberance cases (F+45 and M+45) and decreases by about 1.9% compared to the F+45 case is discussed. Figure 15 shows the local side force distribution in the axial direction of the vehicle. According to this figure, unlike the M+135 and F+45_M+135 cases (Fig. 9), the local side force of the M+45 and F+45_M+45 cases show different tendency behind the Middle protuberance. For the F+45_M+45 case, the local side force increases gradually toward the rear of the body, but for the M+45 case, it remains almost unchanged. We discuss the causes of these differences.

Figure 16 shows the visualization results of the Q iso-surfaces viewed from upstream of the vehicle on the port side, and Fig. 17 shows the velocity vectors and vorticity contours of the $x/L = 0.57$ cross section. According to Fig. 17a, the separation vortex V4 and the horseshoe vortex V6 are generated in the M+45 case. Figure 16a shows that these vortices (V4 and V6) merge with the separation vortex V1, which is generated at the front portion of the slender body at $x/L = 0.8$, to form the vortex V3'. For the F+45_M+45 case, also the horseshoe vortex V6 is generated from the Middle protuberance, as shown in Fig. 17b. Figure 16b shows that this horseshoe vortex (V6) moves farther away from the surface of the vehicle without interfering with the other vortices in the F+45_M+45 case. Figure 18 shows the surface pressure distribution and the visualization results of vorticity at $x/L = 0.8$ cross section to study quantitatively the effect of such a difference in vortex behavior on the vehicle. According to this figure, for the M+45 case, the asymmetry between vortices V1 and V3' is lost, and there is almost no pressure difference between port and starboard sides. Therefore, the local side force is almost zero for $x/L > 0.8$ in Fig. 15. Whereas, in the F+45_M+45 case, vortices V6 and V3 are not in close proximity to the vehicle. Hence, the port side has high pressure than the starboard because the vortex core, which has low pressure, is farther away from the vehicle. This is the reason for the gradual increase in the local side force shown in Fig. 15. (Compared with the F+45 case, the surface pressure on the port side is slightly low because of the presence of vortex V6 near the vehicle (marked "V6" in the graph). This reduces the local side force more than the F+45 case for $0.7 < x/L < 0.9$.)

As described above, even if the Middle protuberance is located at the same location, the vortices originating from the Middle protuberance behave differently downstream of the protuberance (M+45: merge with other vortices, F+45_M+45: does not interfere with other vortices), thus the side force of the F+45_M+45 is not the sum of the single protuberance cases (F+45 and M+45).

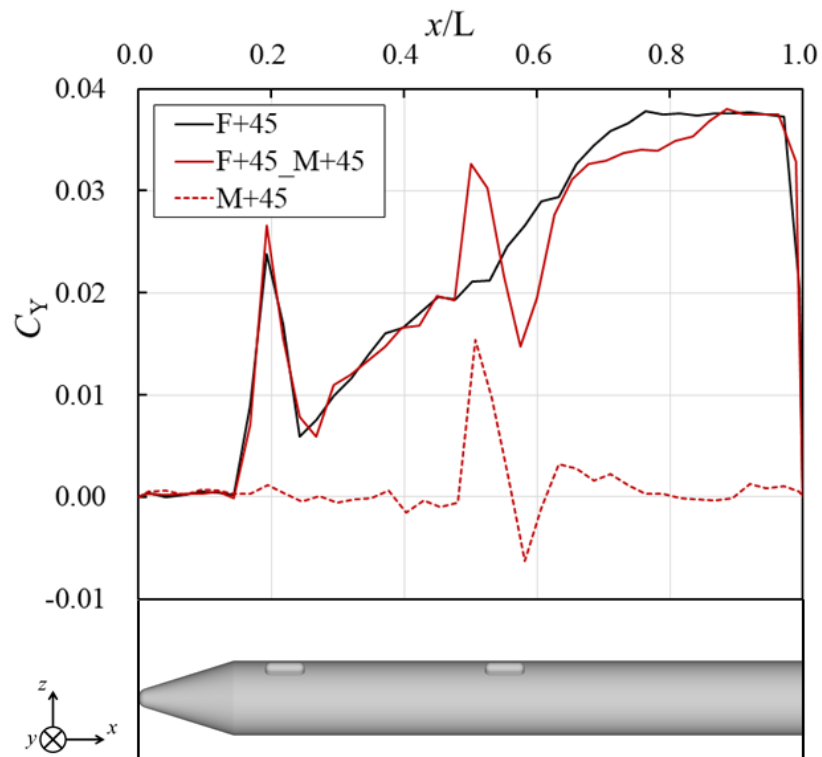


Fig. 15 Local C_Y distribution in the axial direction.

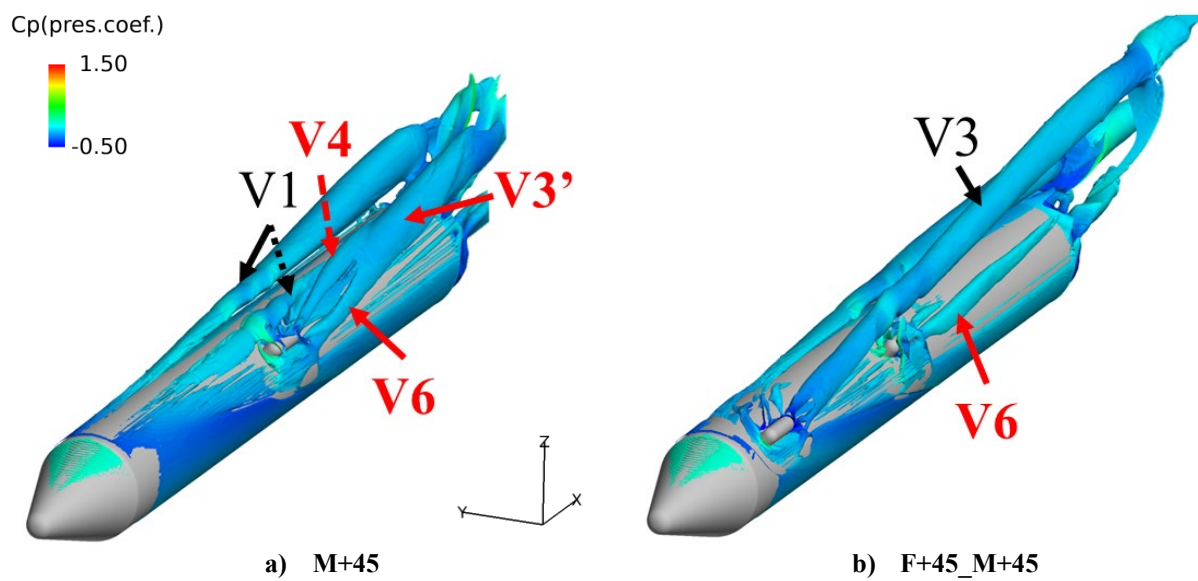


Fig. 16 Q iso-surface (colored with C_p).

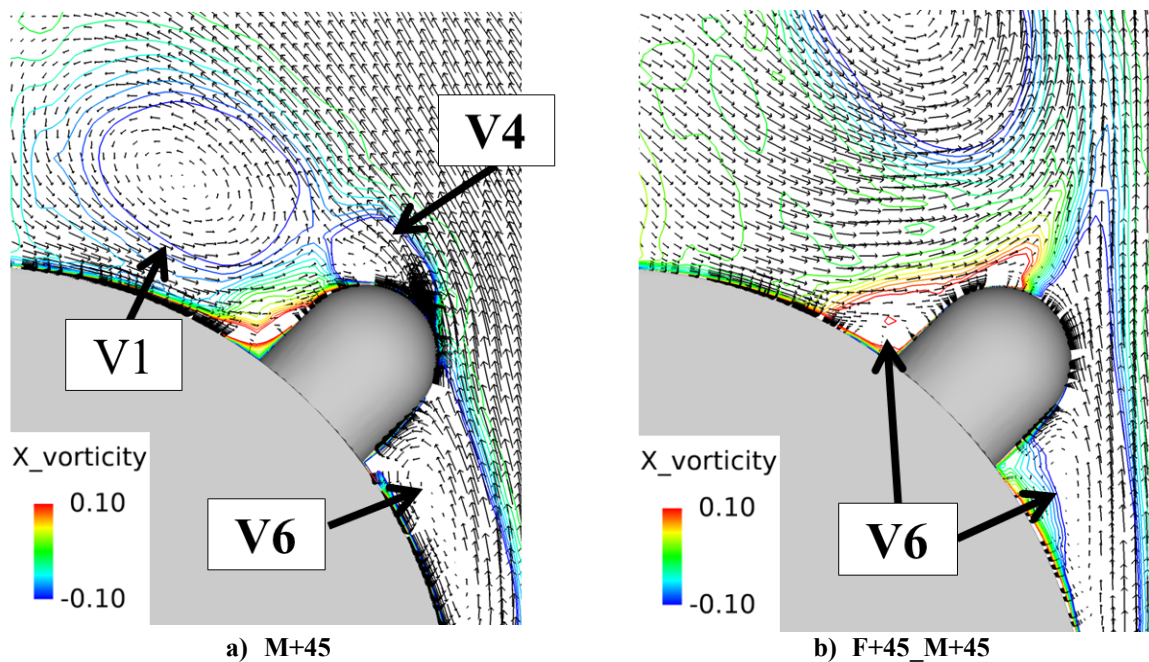


Fig. 17 Velocity vectors and vorticity contours at $x/L = 0.57$ cross section.

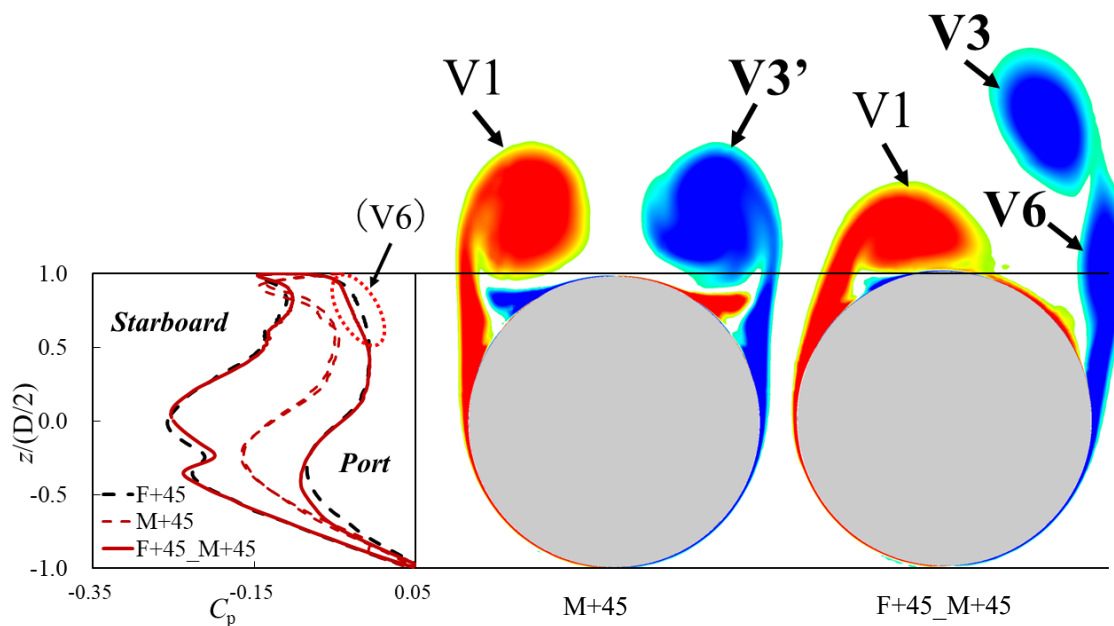


Fig. 18 C_p distributions and two-dimensional (y - z) plane colored with vorticity at $x/L = 0.8$.

IV. Conclusions

In this study, supersonic aerodynamic characteristics of a slender-bodied vehicle with double protuberances, which are mounted asymmetrically, are investigated. We set the positions of the protuberance as follows; i) Front; $x/L = 0.22$, and ii) Middle; $x/L = 0.55$. Azimuthal angle ϕ is set so that the leeward side of the symmetrical plane is 0° , and the windward side is 180° . In the previous study on the single protuberance case, especially large side force from the port side to the starboard side was produced when the angle of attack is 15° with the protuberance installed at an axial position "Front" ($x/L = 0.22$) and an azimuthal angle $\phi = 45^\circ$. Based on this case (F+45), we added another protuberance at "Middle" ($x/L = 0.55$) at azimuthal angles $\phi = 45^\circ, 90^\circ$ and 135° (note that these arrangements are all port side of the body). As for the name of the case, for example, a case in which one protuberance is located at the axial position "Front" and the azimuthal angle $\phi = 45^\circ$, and another at the axial position "Middle" and the azimuthal angle $\phi = 90^\circ$ will be coined as "F+45_M+90." The results are discussed in comparison with the single protuberance cases.

The findings of this study are summarized below.

- Cases F+45_M+135 and M+135:
 - The expansion wave from the anterior part of the Middle protuberance and the separation vortex V4 form the negative pressure region at the leeward of the protuberance. Hence, the local side force is significantly reduced around the Middle protuberance.
 - At the aft portion of the F+45_M+135 case, the local side force is reduced because the surface pressure on the portside decreases as the vortices V4 and V5 are in close proximity to the body surface.
 - The side force of the F+45_M+135 case ($C_Y = 0.787$) is close to the sum of those of the single protuberance cases (F+45; $C_Y = 0.839$ and M+135; $C_Y = -0.059$). This is because the vortices V4 and V5 originating from the Middle protuberance are independent to each other in both M+135 and F+45_M+135 cases, and the local side force distributions at the aft portion of the vehicle in both cases show a similar tendency.
- Cases F+45_M+45 and M+45:
 - The side force of the F+45_M+45 case ($C_Y = 0.823$) is about 1.9% less than that of the F+45 case ($C_Y = 0.839$). The reason for this is that the horseshoe vortex V6 does not interfere with other vortices and exists independently of the other vortices in the wake of the Middle protuberance, reducing the surface pressure on the port side of the vehicle more than the F+45 case, albeit slightly.
 - On the other hand, in the M+45 case, the vortices V4 and V6 originating from the Middle protuberance merge with the separation vortex V1 generated at the front portion of the vehicle. Because of the different behavior of the Middle protuberance-derived vortices in the F+45_M+45 and M+45 cases, the side force of the F+45_M+45 case is not the sum of the single protuberance cases (F+45 and M+45).
- From the above mechanisms, it was found that the side force can be reduced slightly for the F+45_M+45 case and significantly for the F+45_M+135 case compared to the F+45 (single protuberance) case. Therefore, in terms of reducing the side force, these placements of protuberances are optimal for efficient and safe flight.

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